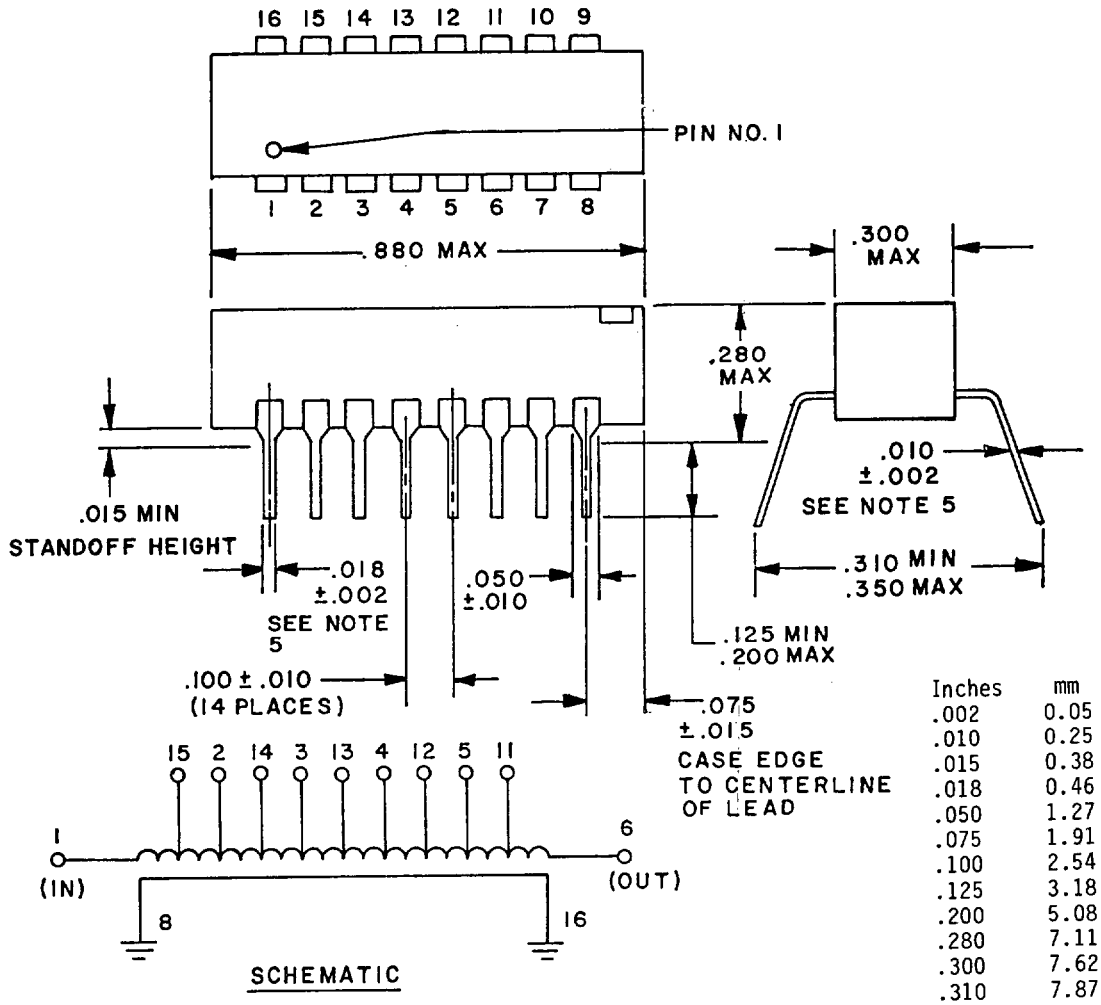
MILITARY SPECIFICATION SHEET

DELAY LINES, 10-TAP, DUAL-IN-LINE, 16-PIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the delay lines described herein shall consist of this specification and the latest issue of MIL-D-83531.



Inches	mm
.002	0.05
.010	0.25
.015	0.38
.018	0.46
.050	1.27
.075	1.91
.100	2.54
.125	3.18
.200	5.08
.280	7.11
.300	7.62
.310	7.87
.350	8.89
.880	22.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Location and shape of standoffs are optional. Standoff height is required.
4. Pins 7, 9 and 10 are not connected.
5. Terminal dimensions are prior to hot solder dip.
6. Schematic is for general information only.

FIGURE 1. Case and mounting dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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REQUIREMENTS:

Operating temperature: -55°C to +125°C.

Weight: 7.5 grams maximum.

Case material: Plastic.

Terminal material: Type LR(0.52) G-35-C* in accordance with MIL-STD-1276. Prior to shipment, terminals shall be hot solder dipped in SN60, SN62, or SN63 in accordance with QQ-S-571 to bottom of standoffs, minimum.

Design and construction: See figure 1.

Total delay time: See table I.

Rise time: 25% of total delay, maximum.

Voltage attenuation: See table I.

Distortion: ±10% at maximum delay.

Delay time variation with temperature: 100 PPM/°C maximum.

DC resistance (RDC): $RDC = \text{impedance}\left(\frac{\text{attenuation}}{1 - \text{attenuation}}\right)$

EXAMPLE: $RDC = 50\left(\frac{0.03}{1-0.03}\right)$ $RDC = 1.5$ ohms

Nominal characteristic impedance: See table I.

Insulation resistance: In accordance with MIL-D-83531.

Input pulse:

Peak pulse voltage: 1 to 3 volts ±5%.

Pulse width: Three times nominal total delay.

Duty cycle: 25%.

Salt spray (corrosion): In accordance with MIL-D-83531.

Thermal shock: In accordance with MIL-D-83531.

Shock: In accordance with MIL-D-83531.

Tap accuracy and total accuracy:

Total delay 0 to 10: ±0.5 ns.
Total delay 11 to 25: ±1 ns or 5% (whichever is greater).
Total delay 26 and up: ±2 ns or 5% (whichever is greater).

Part number: The part number shall be as follows:

EXAMPLE:

M	83531/02	-001
Military designator	Specification sheet number (note zero)	Dash number (see table I)

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TABLE I. Part numbers and electrical measurements.

Part number M83531/02-	Total delay (ns) ±5 percent	Nominal characteristic impedance (Ω) ±10 percent	Maximum attenuation (percent)
001	50	250	2
002	75	250	2
003	100	250	3
004	125	250	3
005	150	250	3
006	200	250	4
007	225	250	4
008	300	250	6
009	375	250	7
010	500	250	9
011	30	300	2
012	60	300	2
013	90	300	2
014	125	300	3
015	150	300	3
016	180	300	4
017	240	300	5
018	270	300	6
019	360	300	7
020	450	300	8
021	600	300	12
022	40	400	2
023	80	400	1
024	120	400	2
025	160	400	4
026	200	400	4
027	240	400	5
028	320	400	6
029	360	400	7
030	480	400	9
031	600	400	10
032	800	400	9
033	200	500	6
034	250	500	3
035	300	500	5
036	400	500	8
037	450	500	8
038	500	500	10
039	600	500	10
040	750	500	9
041	1,000	500	12

1/ Delay between successive taps shall be 1/10 of the nominal total delay time.

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QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83531, for example:

Qualification and testing of M83531/02-001 and M83531/02-041 shall be sufficient to grant qualification to M83531/02-001 through M83531/02-041 inclusive.

Supersession data: MIL-D-83531/2 supersedes DESC drawing 84035 when a qualified source is available.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0170-02)

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